PROGRAM CODE :

```
clc;
clear;
clf();

// Define Laplace variable
s = poly(0, 's');

// Controller parameters
Kp = 5;
Ki = 0.5;
Kd = 2.5;

// Time vector
t = 0:0.01:20;

// First-order process G(s) = 1/(s+1)
Gp = syslin('c', 1/(s+1));

// Controller transfer functions
Gc_P = Kp;
Gc_PI = Kp + Ki/s;
Gc_PID = Kp + Ki/s + Kd*s;

// Closed-loop transfer functions
P = syslin('c', (Gc_P*Gp)/(1+Gc_P*Gp));

PI = syslin('c', (Gc_PI*Gp)/(1+Gc_PI*Gp));
```

```
PID = syslin('c', (Gc_PID*Gp)/(1+Gc_PID*Gp));

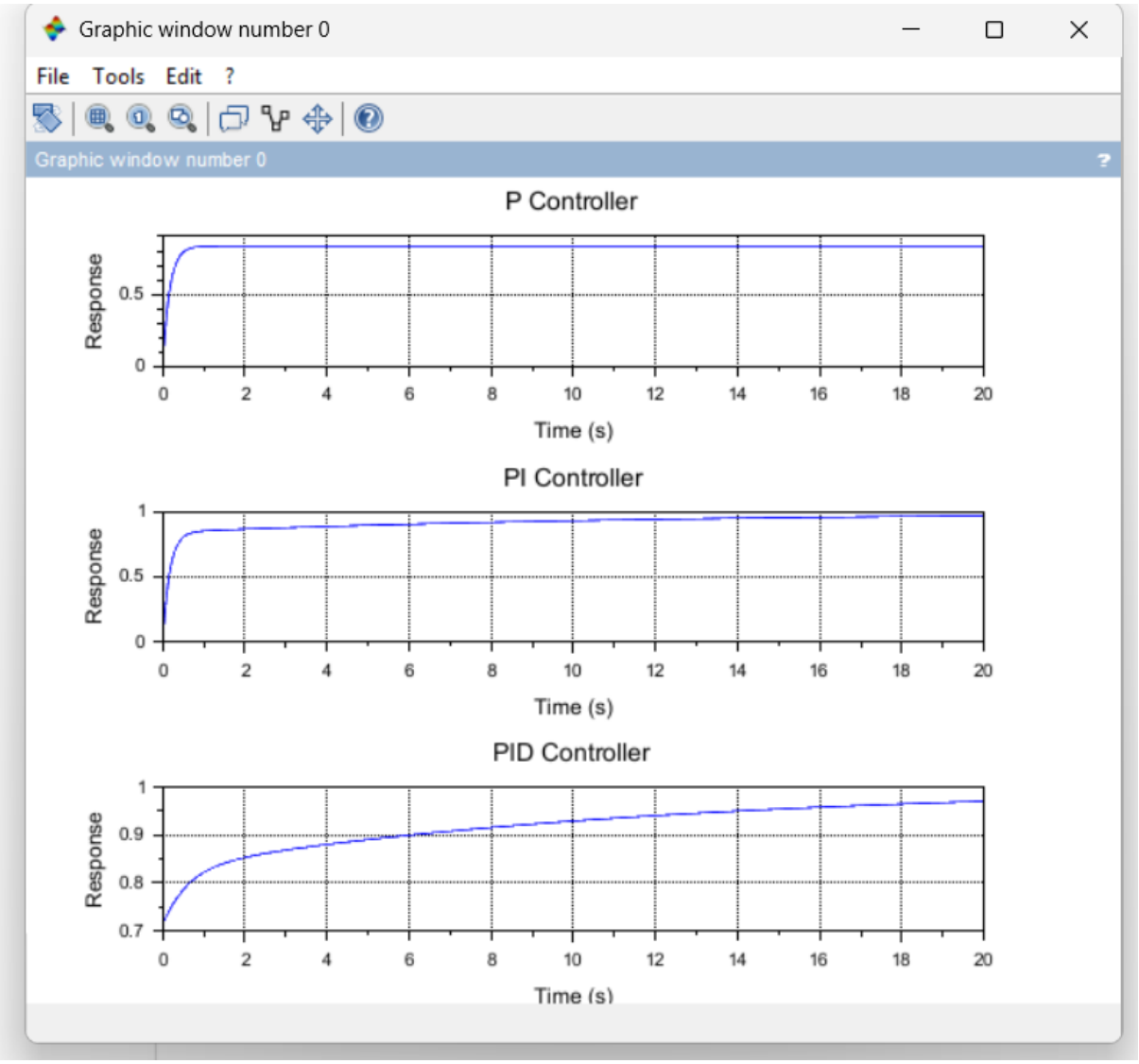
// Step responses
y1 = csim('step', t, P);
y2 = csim('step', t, PI);
y3 = csim('step', t, PID);

// Plot P Controller
subplot(3,1,1);
plot(t,y1);
xtitle('P Controller','Time (s)','Response');
xgrid();

// Plot PI Controller
subplot(3,1,2);
plot(t,y2);
xtitle('PI Controller','Time (s)','Response');
xgrid();

// Plot PID Controller
subplot(3,1,3);
plot(t,y3);
xtitle('PID Controller','Time (s)','Response');
xgrid();
```

OUTPUT :


Graphic window number 0
File Tools Edit ?
Graphic window number 0
P Controller
Response
0
0.5
0 2 4 6 8 10 12 14 16 18 20
Time (s)
PI Controller
Response
0
0.5
1
0 2 4 6 8 10 12 14 16 18 20
Time (s)
PID Controller
Response
0.7
0.8
0.9
1
0 2 4 6 8 10 12 14 16 18 20
Time (s)